Assignment Part 1:
Designing a Database for Commonwealth Transport Services (CTS)

ASSIGNMENT TITLE: ***Logical Database Model for Commonwealth Transport Services***

COURSE CODE(S): 2814ICT

GROUP NUMBER: 35

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	Total contributions	**100%**		

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<table>
<tr><td colspan="3"><u>Declaration</u>
Except where appropriately acknowledged, this assignment is our own work, has been expressed in our own words and has not previously been submitted for assessment. We have also retained a copy of this assessment piece for our own records.</td></tr>
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Date: 20/4/2023</td><td>Name:
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List of Illustrations

Acknowledgements:

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4) Shikha Anirban
5) Tanzima Azad
6) Lutfun Nahar

Entity Relationship Diagram

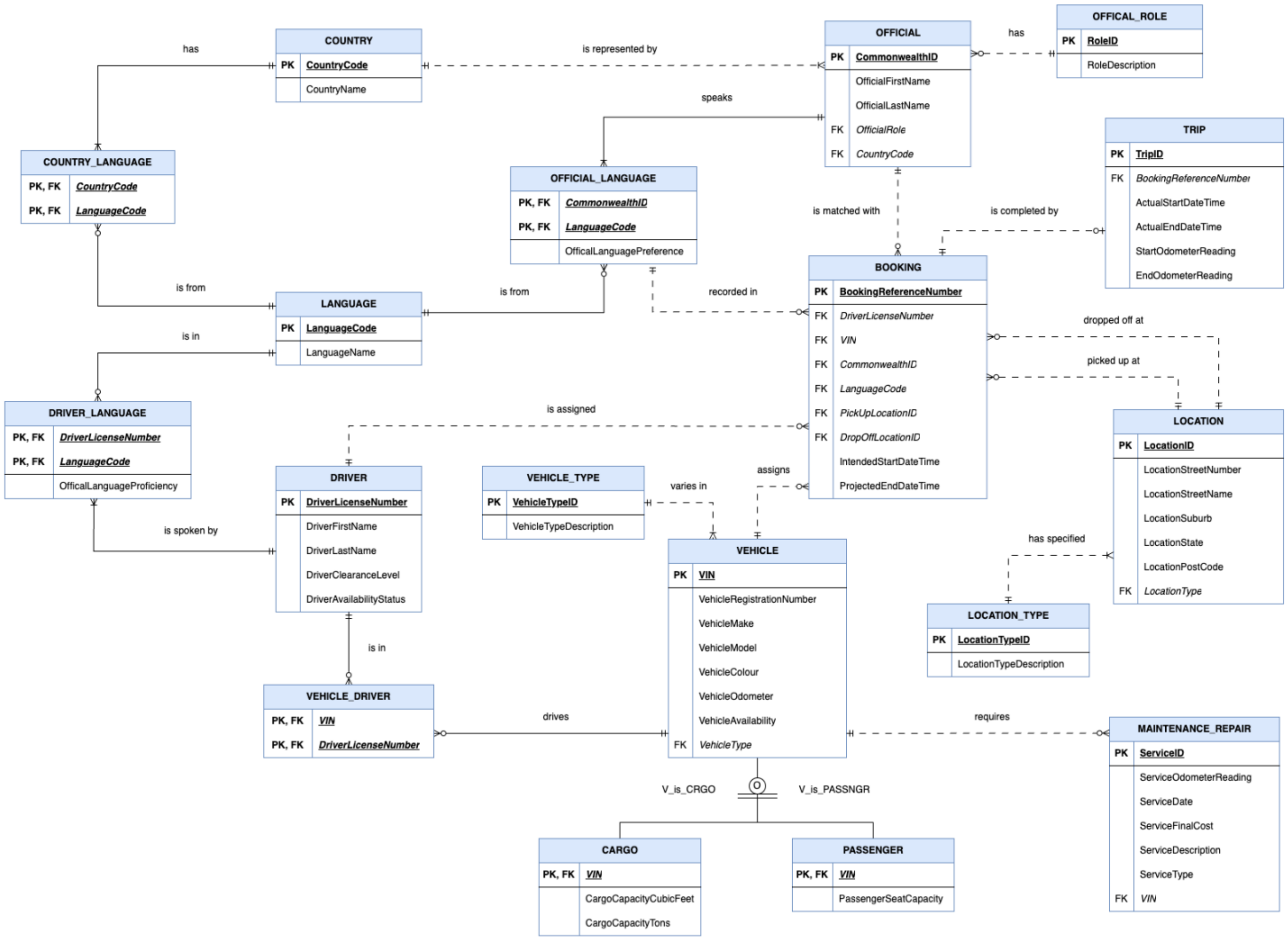


Figure 1: Logical Entity Relationship Diagram

Assumptions

- A language can be spoken in several countries.
- A particular language may not be spoken by any countries.
- A particular language may not be spoken by any drivers or officials.
- A booking only has 1 trip.
- A booking can be cancelled.
- A particular driver and vehicle may not be assigned to a booking.
- An official may not have a booking.
- The “name” attribute of officials and drivers consists of their first name and last name.
- The start time of a trip effects the end time of a trip.
- As there is a “State” attribute, it is assumed these Commonwealth Games are fictitious and were held in Australia.

Normalisation

a) Relation Schema

1. VEHICLE (**<u>VIN</u>**, VehicleRegistrationNumber, VehicleMake, VehicleModel, VehicleColour, VehicleOdometer, VehicleAvailability, *VehicleType*)

2. VEHICLE_TYPE (**<u>VehicleTypeID</u>**, VehicleTypeDescription)

3. CARGO (***<u>VIN</u>***, CargoCapacityCubicFeet, CargoCapacityTons)

4. PASSENGER (***<u>VIN</u>***, PassengerSeatCapacity)

5. MAINTENANCE_REPAIR (**<u>ServiceID</u>**, ServiceOdometerReading, ServiceDate, ServiceFinalCost, ServiceDescription, ServiceType, *VIN*)

6. COUNTRY (**<u>CountryCode</u>**, CountryName)

7. LANGUAGE (**<u>LanguageCode</u>**, LanguageName)

8. COUNTRY_LANGUAGE (***<u>CountryCode</u>***, ***<u>LanguageCode</u>***)

9. OFFICIAL (**<u>CommonwealthID</u>**, OfficialFirstName, OfficialLastName *OfficialRole*, *CountryCode)*

10. OFFICIAL_ROLE (**<u>RoleID</u>**, RoleDescription)

11. OFFICIAL_LANGUAGE (***<u>CommonwealthID</u>***, ***<u>LanguageCode</u>***, OfficialLanguagePreference)

12. DRIVER (**<u>DriverLicenseNumber</u>**, DriverFirstName, DriverLastName, DriverClearanceLevel, DriverAvailabilityStatus)

13. VEHICLE_DRIVER (***<u>VIN</u>***, ***<u>DriverLicenseNumber</u>***)

14. DRIVER_LANGUAGE (***<u>DriverLicenseNumber</u>***, ***<u>LanguageCode</u>***, OfficialLanguageProficiency)

15. BOOKING (**<u>BookingReferenceNumber</u>**, *DriverLicenseNumber*, *VIN*, *CommonwealthID*, *LanguageCode*, *PickUpLocationID*, *DropOffLocationID*, IntendedStartDateTime, IntendedEndDateTime)

16. LOCATION (**<u>LocationID</u>**, LocationStreetNumber, LocationStreetName, LocationSuburb, LocationState, LocationPostCode, *LocationType*)

17. LOCATION_TYPE (**<u>LocationTypeID</u>**, LocationTypeDescription)

18. TRIP (**<u>TripID</u>**, *BookingReferenceNumber*, ActualStartDateTime, ActualEndDateTime, StartOdometerReading, EndOdometerReading)

b) Normalisation

1. Dependency Table for **VEHICLE**:

Full-Functional

VEHICLE	**<u>VIN</u>**	VehicleRegistrationNumber	VehicleMake	VehicleModel	VehicleColour	VehicleOdometer	VehicleAvailability	*VehicleType*

Transitive

This table is in 2NF due to the absence of any partial dependencies and the presence of transitive and full-functional dependencies.

The aforementioned transitive dependency exists between the VehicleMake and VehicleModel attributes. The specific make (brand) of a vehicle is determined by the particular vehicle model therefore, a transitive dependency is evident between these particular attributes. However, the VehicleMake attribute introduces negligible redundancy and thus, decomposition into 3NF is not required.

2. Dependency Table for **VEHICLE_TYPE**:

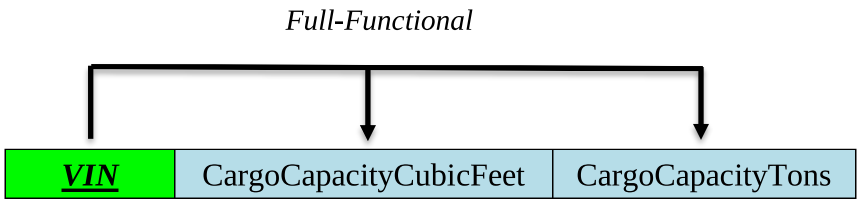
Full-Functional

VEHICLE_TYPE	**<u>VehicleTypeID</u>**	VehicleTypeDescription

This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

3. Dependency Table for **CARGO:**

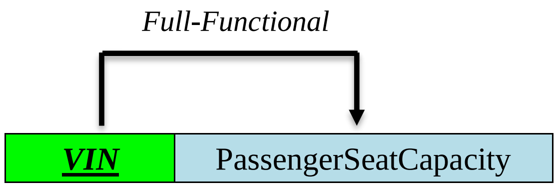
CARGO



This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

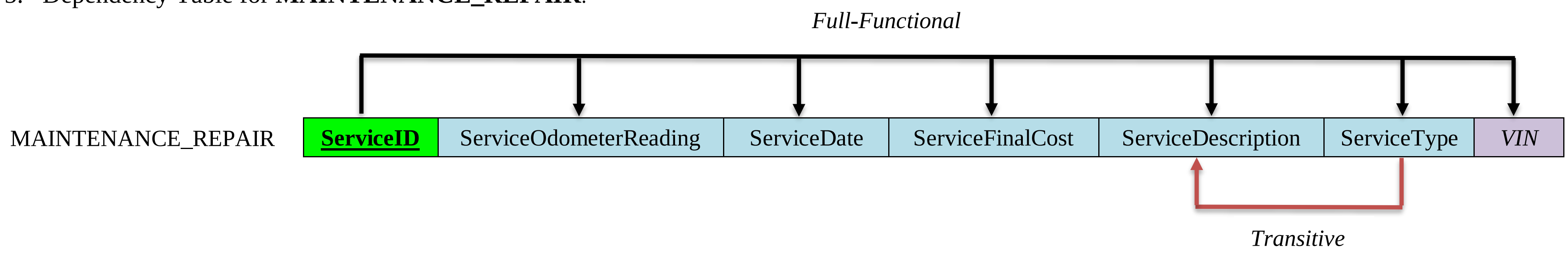
4. Dependency Table for **PASSENGER:**

PASSENGER



This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

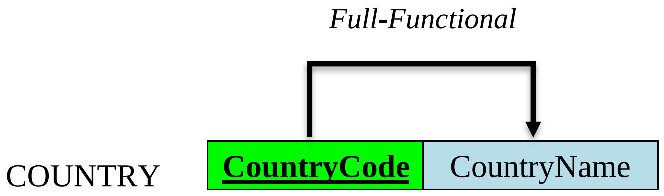
5. Dependency Table for **MAINTENANCE_REPAIR**:



This table is in 2NF due to the absence of any partial dependencies and the presence of transitive and full-functional dependencies.

The aforementioned transitive dependency exists between the ServiceType and ServiceDescription attributes. The description associated to a given vehicle service is dependent on the type of service conducted for instance, maintenance or repair therefore, a transitive dependency is evident between these particular attributes. However, the ServiceDescription attribute introduces negligible redundancy and thus, decomposition into 3NF is not required.

6. Dependency Table for **COUNTRY**:



This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

7. Dependency Table for **LANGUAGE**:

Full-Functional

LANGUAGE

<u>LanguageCode</u>	LanguageName

This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

8. Dependency Table for **COUNTRY_LANGUAGE**:

Full-Functional

COUNTRY_LANGUAGE

<u>CountryCode</u>	***<u>LanguageCode</u>***

This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

9. Dependency Table for **OFFICIAL:**

Full-Functional

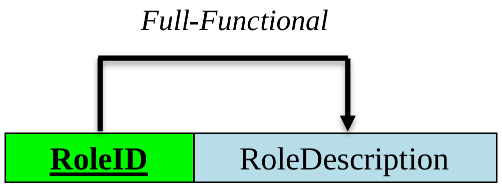
OFFICIAL

<u>CommonwealthID</u>	OfficialFirstName	OfficialLastName	*OfficialRole*	*CountryCode*

This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

10. Dependency Table for **OFFICAL_ROLE**:

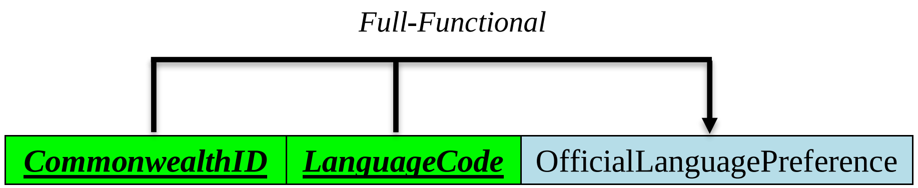
OFFICAL_ROLE



This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

11. Dependency Table for **OFFICIAL_LANGUAGE:**

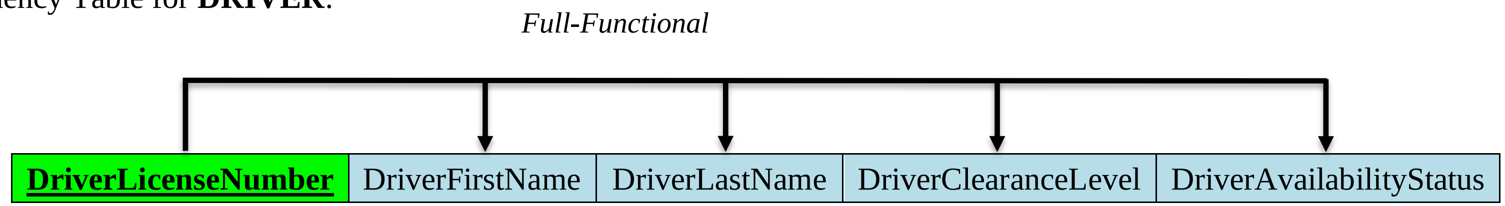
OFFICIAL_LANGUAGE



This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

12. Dependency Table for **DRIVER**:

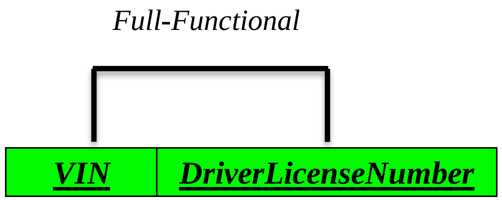
DRIVER



This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

13. Dependency Table for **VEHICLE_DRIVER**:

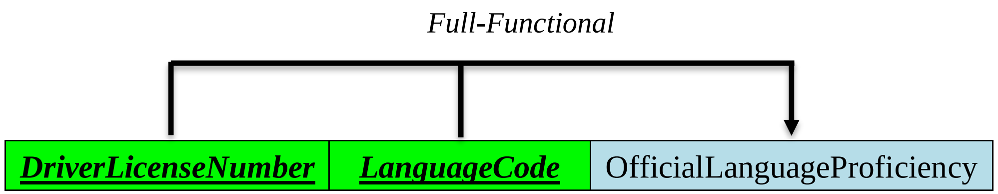
VEHICLE_DRIVER



This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

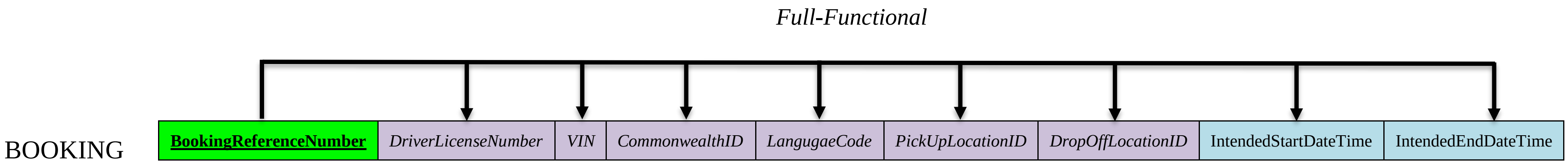
14. Dependency Table for **DRIVER_LANGUAGE:**

DRIVER_LANGUAGE



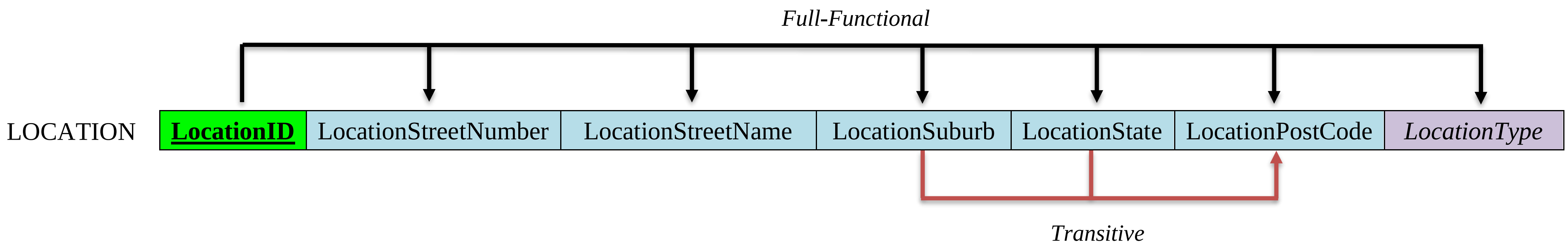
This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

15. Dependency Table for **BOOKING:**



This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

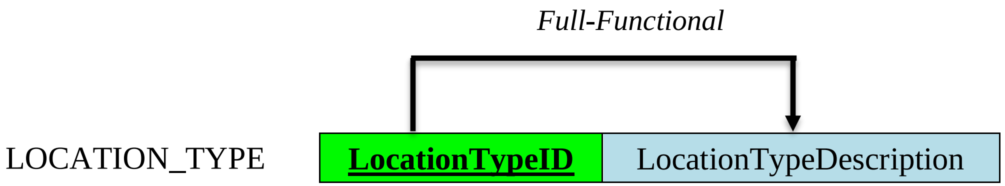
16. Dependency Table for **LOCATION:**



This table is in 2NF due to the absence of any partial dependencies and the presence of transitive and full-functional dependencies.

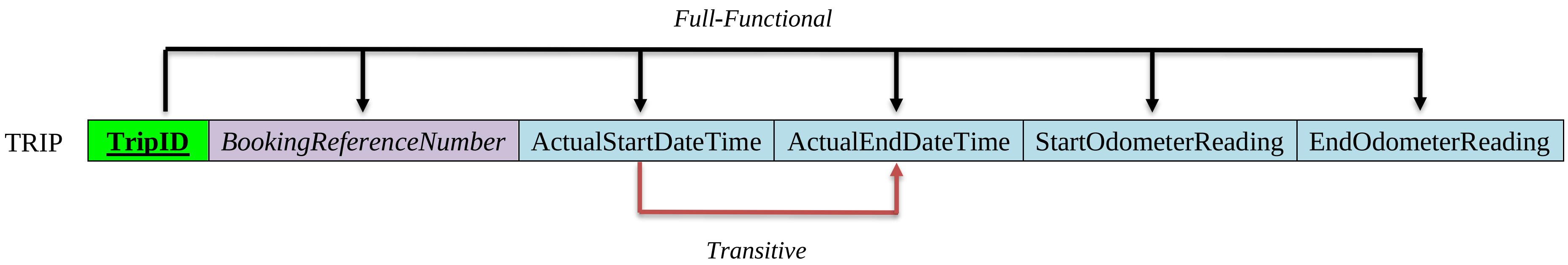
The aforementioned transitive dependency exists between LocationState, LocationSuburb and the subsequent LocationPostCode. A specific postcode is determined by a combination of the suburb and state the particular area is located in therefore, a transitive dependency is evident between these attributes. However, the Postcode attribute introduces negligible redundancy and thus, decomposition into 3NF is not required.

17. Dependency Table for **LOCATION_TYPE**:



This table is in 3NF due to the absence of any transitive or partial dependencies and the presence of full-functional dependencies throughout.

18. Dependency Table for **TRIP:**



This table is in 2NF due to the absence of any partial dependencies and the presence of transitive and full-functional dependencies.

The aforementioned transitive dependency exists between the ActualStartDateTime and ActualEndDateTime attributes. The time of completion for a specific trip is dependent on the time it was commenced; thus, a transitive dependency is evident between these particular attributes. Nonetheless, the ActualStartDateTime attribute introdu mces negligible redundancy and thus, decomposition into 3NF is not required.

Relational Database Schema

Relation Name	Attribute	Data Type	Length	Description
VEHICLE	VIN	CHAR	17	PRIMARY KEY
	VehicleRegistrationNumber [1]	VARCHAR	7	
	VehicleMake	VARCHAR	30	
	VehicleModel	VARCHAR	30	
	VehicleColour	VARCHAR	20	
	VehicleOdometer [2]	INT	6	
	VehicleAvailability	CHAR	1	
	VehicleType	CHAR	2	FOREIGN KEY REFERENCES VEHICLE_TYPE (VehicleTypeID)
VEHICLE_TYPE	VehicleTypeID	CHAR	2	PRIMARY KEY
	VehicleTypeDescription	VARCHAR	40	
CARGO	VIN	CHAR	17	PRIMARY KEY, FOREIGN KEY REFERENCES VEHICLE (VIN)
	CargoCapacityCubicFeet [3]	NUMERIC	4, 2	
	CargoCapacityTons [3]	NUMERIC	2, 1	
PASSENGER	VIN	CHAR	17	PRIMARY KEY, FOREIGN KEY REFERENCES VEHICLE (VIN)
	PassengerSeatCapacity	TINYINT	2	
MAINTENANCE_REPAIR	ServiceID	INT (AUTO)		PRIMARY KEY
	ServiceOdometerReading	INT	6	
	ServiceDate	DATE		**FORMAT:** DD-MM-YYYY
	ServiceFinalCost	NUMERIC	5, 2	
	ServiceDescription	VARCHAR	100	
	ServiceType	CHAR	1	Maintenance (M) or Repair (R)
	VIN	CHAR	17	FOREIGN KEY REFERENCES VEHICLE (VIN)
COUNTRY	CountryCode	CHAR	2	PRIMARY KEY
	CountryName	VARCHAR	50	
LANGUAGE	LanguageCode	CHAR	2	PRIMARY KEY
	LanguageName	VARCHAR	50	
COUNTRY_LANGUAGE	CountryCode	CHAR	2	PRIMARY KEY, FOREIGN KEY REFERENCES COUNTRY (CountryCode)
	LanguageCode	CHAR	2	PRIMARY KEY, FOREIGN KEY REFERENCES LANGUAGE (LanguageCode)
OFFICIAL	CommonwealthID	CHAR	8	PRIMARY KEY
	OfficialFirstName	VARCHAR	40	
	OfficialLastName	VARCHAR	40	
	OfficialRole	CHAR	2	FOREIGN KEY REFERENCES OFFICAL_ROLE (RoleID)
	CountryCode	CHAR	2	FOREIGN KEY REFERENCES COUNTRY (CountryCode)
OFFICIAL_ROLE	RoleID	CHAR	2	PRIMARY KEY
	RoleDescription	VARCHAR	40	
OFFICIAL_LANGUAGE	CommonwealthID	CHAR	8	PRIMARY KEY, FOREIGN KEY REFERENCES OFFICIAL (CommonwealthID)
	LanguageCode	CHAR	2	PRIMARY KEY, FOREIGN KEY REFERENCES LANGUAGE (LanguageCode)
	OfficialLanguagePreference	TINYINT	2	
DRIVER	DriverLicenseNumber	CHAR	18	PRIMARY KEY
	DriverFirstName	VARCHAR	40	
	DriverLastName	VARCHAR	40	
	DriverClearanceLevel	TINYINT	1	
	DriverAvailabilityStatus	CHAR	1	

VEHICLE_DRIVER	VIN	CHAR	17	PRIMARY KEY, FOREIGN KEY REFERENCES VEHICLE (VIN)
	DriverLicenseNumber	CHAR	18	PRIMARY KEY, FOREIGN KEY REFERENCES DRIVER (DriverLicenseNumber)
DRIVER_LANGUAGE	DriverLicenseNumber	CHAR	18	PRIMARY KEY, FOREIGN KEY REFERENCES DRIVER (DriverLicenseNumber)
	LanguageCode	CHAR	2	PRIMARY KEY, FOREIGN KEY REFERENCES LANGUAGE (LanguageCode)
	DriverLanguageProficiency [4]	VARCHAR	40	
BOOKING	BookingReferenceNumber	INT (AUTO)		PRIMARY KEY
	DriverLicenseNumber	CHAR	18	FOREIGN KEY REFERENCES DRIVER (DriverLicenseNumber)
	VIN	CHAR	17	FOREIGN KEY REFERENCES VEHICLE (VIN)
	CommonwealthID	CHAR	2	FOREIGN KEY REFERENCES OFFICIAL_LANGUAGE (CommonwealthID)
	LanguageCode	CHAR	2	FOREIGN KEY REFERENCES OFFICAL_LANGUAGE (LanguageCode)
	PickupLocationID	INT		FOREIGN KEY REFERENCES LOCATION (LocationID)
	DropOffLocationID	INT		FOREIGN KEY REFERENCES LOCATION (LocationID)
	IntendedStartDateTime	DATETIME		**FORMAT:** DD-MM-YYYY HH:MM:SS
	ProjectedEndDateTime	DATETIME		**FORMAT:** DD-MM-YYYY HH:MM:SS
LOCATION	LocationID	INT (AUTO)		PRIMARY KEY
	LocationStreetNumber	SMALLINT	5	
	LocationStreetName	VARCHAR	50	
	LocationSuburb	CHAR	50	
	LocationState	CHAR	3	
	LocationPostCode	CHAR	4	
	LocationType	CHAR	2	FOREIGN KEY REFERENCES LOCATION_TYPE (LocationTypeID)
LOCATION_TYPE	LocationTypeID	CHAR	2	PRIMARY KEY
	LocationTypeDescription	VARCHAR	40	
TRIP	TripID	INT (AUTO)		PRIMARY KEY
	BookingReferenceNumber	INT		FOREIGN KEY REFERENCES BOOKING (BookingReferenceNumber)
	ActualStartDateTime	DATETIME		**FORMAT:** DD-MM-YYYY HH:MM:SS
	ActualEndDateTime	DATETIME		**FORMAT:** DD-MM-YYYY HH:MM:SS
	StartOdometerReading	INT	6	
	EndOdometerReading	INT	6	

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